

Example 4: Let $Q(x, y)$ denote " $x+y=0$." What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Soln: The quantification $\exists y \forall x Q(x, y)$ denotes the proposition "There is a real number y such that for every real number x , $Q(x, y)$." No matter what value of y is chosen, there is only one value of x for which $x+y=0$. Because there is no real number y such that $x+y=0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition "For every real number x there is a real number y such that $Q(x, y)$." Given a real number x , there is a real number y such that $x+y=0$; namely, $y=-x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true $\square$

Example 4 illustrates that the order in which quantifiers appear make a difference. The statements $\exists y \forall x P(x, y)$ and $\forall x \exists y P(x, y)$ are not logically equivalent. The statement $\exists y \forall x P(x, y)$ is true if and only if there is a y that makes $P(x, y)$ true for every x . So, for this statement to be true, there must be a particular value of y for which $P(x, y)$ is true regardless of the choice of x . On the other hand, $\forall x \exists y P(x, y)$ is true if and only if for every value of x there is a value of y for which $P(x, y)$ is true. So for this statement to be true, no matter which x you choose, there must be a value of y (possibly depending on the x you choose) for which $P(x, y)$ is true. In other words, in the second case, y can depend on x , whereas in the first case, y is a constant independent of x .